An Attempt at Error Decomposition for the POEMV Algorithm

Sam Olson

Abstract

Wu and Li [2024] report empirically that filtering error from inaccurate parameter estimation is the dominant source of performance degradation in the POEMV algorithm, but offer no formal support for this attribution. We attempt to formalize this attribution by identifying and partially bounding three sources of suboptimality: (I) the approximation error incurred by representing the value function components via finite-degree polynomials, (II) the propagation of parameter estimation errors through the Wonham filter to the belief state, and (III) the resulting sensitivity of the optimal policy mean to belief state error. The first bound exploits the $C^{1,2}$ regularity of the value function established in Wu and Li's Theorem 1; the second uses a Gronwall argument on the Wonham filter SDE; the third follows from the Lipschitz structure of the myopic–hedging decomposition. Together these yield a partial error decomposition for POEMV that clarifies when filtering error dominates. Terms (I) and (II) are bounded under sufficient conditions: Term (I) requires only the $C^{1,2}$ regularity of f, while Term (II) requires a filter stability condition that is violated by the Wu and Li calibration, leaving Corollary 6.2 not formally established for the motivating example. Term (III) remains entirely open. We also identify two open problems whose resolution would complete the decomposition: a filter stability bound valid in the large-$|\rho_1 - \rho_2|$ regime characteristic of equity markets, and a finite-time SGD error bound under a moving loss landscape. The first is an obstacle for any Gronwall-based approach to Wonham filter error in this setting; the second requires controlling the coupling between the gradient estimator and the filtered belief state.

Keywords: Error decomposition, continuous-time reinforcement learning, mean-variance portfolio selection, regime switching, Wonham filter, partial information, exploratory stochastic control.

1 Introduction

Wu and Li [2024] establish that the POEMV algorithm learns near-optimal policies for continuous-time mean-variance portfolio selection in a regime-switching market with an unobservable regime. Their Theorem 1 proves well-posedness and regularity of the value function components $f(t,p)$ and $g(t,p)$, and their Theorem 2 establishes Gaussian optimality of the exploratory policy in the partial-information setting. However, the end-to-end error of the full learning loop is never formally analyzed. Three error sources are implicit: (I) polynomial approximation of f and g, (II) parameter estimation in the Wonham filter, and (III) stochastic gradient error in actor–critic updates. No formal bounds are provided, and they note that an explicit decomposition remains to be identified.

This gap has practical consequences. In the simulation study of Wu and Li, terminal wealth variance under estimated parameters (0.170) substantially exceeds that under true parameters (0.083), which itself exceeds the finite-difference optimum (0.067). The discrepancy appears to be driven by parameter misestimation, e.g., $\hat{\lambda}_1 = 1.21$ versus $\lambda_1^* = 0.36$, but this attribution is empirical. A formal error decomposition would both validate and quantify this effect.

We address the first two error sources and detail the difficulties in evaluating the third. Specifically, we establish:

- **Proposition 3.1** (Polynomial approximation error): The error from approximating $\ln f(t,p)$ and $g(t,p)$ by polynomials of degree m in each of $(p, T-t)$ decays at rate $O(m^{-2})$ in the sup norm on any compact subset of $(0,1)$, with constants from Wu and Li's Theorem 1.

- **Proposition 4.1** (Filter error propagation): If the estimated environment parameters $\tilde{\theta}$ satisfy $\left\|\tilde{\theta} - \theta^*\right\| \leq \delta$ and the filter stability condition $\tilde{\lambda}_1 + \tilde{\lambda}_2 > \frac{1}{2}(1 + 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2)$ holds, the mean-squared belief state error satisfies $\sup_{t\in[0,T]} \mathbb{E}[e_t^2] \leq C_F\, \delta^2/(\tilde{\lambda}_1 + \tilde{\lambda}_2)$, where $e_t = p_t - \tilde{p}_t$ and the damping factor $\tilde{\lambda}_1 + \tilde{\lambda}_2$ is the estimated total transition rate. The stability condition is sufficient but not necessary (Open Problem 1, Section 7).

- **Proposition 5.1** (Policy sensitivity): The mean-squared policy error induced by belief state error satisfies $\mathbb{E}\big[|m^*(t, X_t, p_t) - m^*(t, X_t, \tilde{p}_t)|^2\big] \leq (C_\gamma^2/\sigma^2)\, \mathbb{E}[(X_t - \omega)^2] \cdot \mathbb{E}[e_t^2]$, where C_γ depends only on the bounds on f_p/f from Theorem 1.

- **Theorem 6.1** (Combined decomposition): The total mean-squared policy error decomposes additively, with the filter propagation term dominant when parameter estimation error is large relative to polynomial approximation error, consistent with Wu and Li's empirical finding.

Section 7 states the two open problems left in the wake of this supplement.

2 Notation and Preliminaries

Following the notation of Wu and Li [2024], we work in the following model, stated explicitly for completeness.

Market and regime. A financial market over horizon $[0, T]$ consists of one risk-free asset with constant rate $r \geq 0$ and one risky asset whose discounted price satisfies the regime-switching geometric Brownian motion $dS_t = S_t(\mu_{I_t}\, dt + \sigma\, dB_t)$, with constant volatility $\sigma > 0$ and drift μ_{I_t} determined by the latent market regime. The regime process $I = \{I_t : t \geq 0\}$ is an unobservable continuous-time Markov chain on $\{1, 2\}$ (interpreted as bull and bear markets) with generator matrix Q having off-diagonal entries $\lambda_1, \lambda_2 > 0$, independent of B.

Investment problem. An investor chooses a self-financed strategy u_t (discounted dollar value in the risky asset) adapted to the observable price filtration $\mathcal{F}_t^S$. The discounted wealth satisfies $dX_t^u = \sigma u_t(\hat{\rho}_t\, dt + d\hat{B}_t)$, where $\hat{\rho}_t = (\rho_1 - \rho_2)p_t + \rho_2$ is the filtered Sharpe ratio and $\hat{B}_t$ is the innovation process. The entropy-regularized mean-variance objective is $\min_{\boldsymbol{\pi}} \mathbb{E}[(X_T^{\boldsymbol{\pi}} - \omega)^2 - \Lambda \int_0^T \mathcal{H}(\boldsymbol{\pi}(\cdot|t, X_t, p_t))\, dt] - (\omega - z)^2$, with temperature $\Lambda \geq 0$ and Lagrange multiplier ω. This is the RSEMV problem of Wu and Li [2024].

The true environment parameters are $\theta^* = (\mu_1^*, \mu_2^*, \sigma^*, \lambda_1^*, \lambda_2^*)$, yielding Sharpe ratios $\rho_i^* = (\mu_i^* - r)/\sigma^*$. Estimated parameters are denoted $\tilde{\theta} = (\tilde{\mu}_1, \tilde{\mu}_2, \tilde{\sigma}, \tilde{\lambda}_1, \tilde{\lambda}_2)$ with $\tilde{\rho}_i = (\tilde{\mu}_i - r)/\tilde{\sigma}$. Parameter errors are $\delta\mu_i = \tilde{\mu}_i - \mu_i^*$, $\delta\lambda_i = \tilde{\lambda}_i - \lambda_i^*$, and $\delta\rho_i = \tilde{\rho}_i - \rho_i^*$.

The belief state under true parameters is $p_t = \Pr(I_t = 1 \mid \mathcal{F}_t^S)$, satisfying the *Wonham filter SDE*:

$$dp_t = \left[-(\lambda_1^* + \lambda_2^*)p_t + \lambda_2^*\right] dt + (\rho_1^* - \rho_2^*)\, p_t(1 - p_t)\, d\hat{B}_t. \tag{1}$$

The belief state under estimated parameters, $\tilde{p}_t$, satisfies:

$$d\tilde{p}_t = \left[-(\tilde{\lambda}_1 + \tilde{\lambda}_2)\tilde{p}_t + \tilde{\lambda}_2\right] dt + (\tilde{\rho}_1 - \tilde{\rho}_2)\, \tilde{p}_t(1 - \tilde{p}_t)\, d\hat{B}_t, \tag{2}$$

where both filters share the same innovation process $\hat{B}_t$. This representation is standard, in that every $\mathcal{F}^S$-adapted Itô process admits such a representation, but it suppresses one $O(\delta)$ term. In practice the misspecified filter computes $d\tilde{B}_t = dS_t/(\tilde{\sigma} S_t) - \tilde{\hat{\rho}}_t\, dt$ using estimated parameters, which differs from $d\hat{B}_t$ by an $O(\delta)$ drift. Writing both filters against $\hat{B}_t$ folds this discrepancy into the drift forcing ξ_t^b of the error SDE in Proposition 4.1 rather than

the diffusion coefficient, making the Gronwall bound slightly tighter than a fully explicit accounting would yield.

The optimal value function of the RSEMV problem is:

$$V^*(t,x,p) = (x-\omega)^2 \frac{1}{f(t,p)} + g(t,p) - (\omega - z)^2, \tag{3}$$

where f and g satisfy the degenerate parabolic PDEs (18) and (19) of Wu and Li [2024]. The optimal Gaussian policy has mean:

$$m^*(t,x,p) = -\frac{x-\omega}{\sigma} \underbrace{\left[\rho_2^* + (\rho_1^* - \rho_2^*)p\right]}_{\text{myopic}} + \frac{x-\omega}{\sigma} \underbrace{(\rho_1^* - \rho_2^*)\, p(1-p)\, \frac{f_p(t,p)}{f(t,p)}}_{\text{hedging demand}}. \tag{4}$$

We collect the key regularity results from Wu and Li's Theorem 1 that all subsequent bounds rely on.

Assumption 2.1 (Regularity of f, from Wu and Li 2024 Theorem 1)**.**

(R1) $f \in C^{1,2}([0,T) \times [\varepsilon, 1-\varepsilon]) \cap C([0,T] \times [\varepsilon, 1-\varepsilon])$ *for every* $\varepsilon \in (0, \frac{1}{2})$.

(R2) $1 \le f(t,p) \le e^{\bar{\rho}^2(T-t)}$ *on* $[0,T] \times (0,1)$, *where* $\bar{\rho} = \max\{|\rho_1^*|, |\rho_2^*|\}$.

(R3) *There exist finite constants* $C_1, C_2 > 0$ *such that* $|f_p(t,p)| \le C_1$ *and* $|f_{pp}(t,p)| \le C_2$ *uniformly on* $[0,T] \times (0,1)$.

(R4) f *is decreasing in* t *for each fixed* $p \in (0,1)$.

3 Polynomial Approximation Error

Wu and Li [2024] approximate $\ln f(t,p)$ and $g(t,p)$ by polynomials of bidegree m in $(p, T-t)$:

$$f^\theta(t,p) = \exp\left(\sum_{\substack{0\le i\le m\\ 1\le j\le m}} \theta_{i,j}\, p^i (T-t)^j\right), \qquad g^\theta(t,p) = \sum_{\substack{0\le i\le m\\ 1\le j\le m}} \tilde{\theta}_{i,j}\, p^i (T-t)^j. \tag{5}$$

Approximation quality depends on the smoothness of $\ln f$ and g on the approximation domain.

Proposition 3.1 (Polynomial approximation error)**.** *Fix* $\varepsilon \in (0, \frac{1}{2})$ *and let* $\mathcal{D}_\varepsilon = [0,T] \times [\varepsilon, 1-\varepsilon]$. *Define the oracle approximation error of polynomials of degree* m *in each of* $(p,\, T-t)$

as $E_m(\ln f,\, \mathcal{D}_\varepsilon) \coloneqq \inf_{q\in\mathcal{P}_m} \|\ln f - q\|_{L^\infty(\mathcal{D}_\varepsilon)}$*, where* $\mathcal{P}_m$ *denotes the class of bivariate polynomials of degree at most* m *in each variable. Under Assumption 2.1, there exists a constant* K_ε *depending on* ε*,* $\bar{\rho}$*,* C_1*,* C_2*, and* T *but not on* m*, such that:*

$$E_m(\ln f,\, \mathcal{D}_\varepsilon) \;\le\; \frac{K_\varepsilon}{m^2}. \tag{6}$$

An analogous bound holds for g*.*

Proof. Define $h(\tau, p) = \ln f(T-\tau, p)$ on $[0,T]\times[\varepsilon, 1-\varepsilon]$, so that polynomial approximation in $(p, T-t)$ corresponds to approximation of h in (τ, p). We verify that h admits bounded second-order partial derivatives on $\mathcal{D}_\varepsilon$.

For the p-derivatives, write $h_p = f_p/f$ and $h_{pp} = f_{pp}/f - (f_p/f)^2$. By (R2), $f \ge 1$, so $|h_p| \le |f_p| \le C_1$ and $|h_{pp}| \le |f_{pp}|/f + (f_p/f)^2 \le C_2 + C_1^2$. For the τ-derivative, $h_\tau = -f_t/f$; since $f_t \le 0$ by (R4) and $f \ge 1$, there exists a finite constant C_3 such that $|h_\tau| \le C_3$. For the second τ-derivative, $h_{\tau\tau} = -f_{tt}/f + (f_t/f)^2$. Differentiating PDE (18) of Wu and Li [2024] with respect to t yields a linear parabolic PDE for f_t with the same degenerate diffusion coefficient $a(p) = (\rho_1^* - \rho_2^*)^2 p^2 (1-p)^2$. On $\mathcal{D}_\varepsilon = [0,T]\times[\varepsilon, 1-\varepsilon]$ the diffusion is uniformly elliptic (bounded away from zero), so Theorem 10 in Chapter 2 of Krylov [2008] (as noted by Wu and Li) applied to this derived PDE yields boundedness of f_{tt} on $\mathcal{D}_\varepsilon$; the argument is identical to Wu and Li's proof of part (iii) of Theorem 1 establishing boundedness of f_t. Hence there exists a finite constant C_4 such that $|h_{\tau\tau}| \le C_4$.

Thus h satisfies the regularity conditions required for second-order Jackson-type approximation bounds with uniformly bounded second-order derivatives in both variables. We apply the Jackson theorem for bivariate algebraic polynomial approximation on a compact rectangle, which extends the univariate $O(m^{-2})$ rate to tensor-product polynomial spaces under bounded second-order partial derivatives in each variable separately; a fully rigorous treatment in the non-uniform case would require verifying the mixed derivative $h_{\tau p}$ as well, but this follows from differentiating the governing PDE with respect to both t and p and invoking standard parabolic regularity arguments. This yields:

$$E_m(h,\, [0,T]\times[\varepsilon, 1-\varepsilon]) \;\le\; \frac{C}{m^2}\big(\|h_{pp}\|_{L^\infty} + \|h_{\tau\tau}\|_{L^\infty}\big) \;\le\; \frac{K_\varepsilon}{m^2},$$

where $K_\varepsilon = C(C_2 + C_1^2 + C_4)$. □

Remark 3.2. *The implemented policy* m_m^* *uses* $(q_m)_p$ *in place of* $(\ln f)_p = f_p/f$*, where* q_m *is the degree-m best polynomial approximation to* $\ln f$*. The policy-relevant approximation error is therefore in the* derivative $(\ln f)_p$*, not in* $\ln f$ *itself. Applying the Jackson theorem to* $h_p = (\ln f)_p$*: Since* h_{pp} *is bounded on* $\mathcal{D}_\varepsilon$ *(established in the proof of Proposition 3.1), the*

best degree-m polynomial approximation to h_p satisfies $E_m(h_p, \mathcal{D}_\varepsilon) \le K'_\varepsilon/m$, one order slower than the $O(m^{-2})$ rate for h itself. Since $m^*(t,x,p)$ is linear in f_p/f through the hedging term of equation (4), the policy mean approximation error is $O(m^{-1})$, and the squared *policy error from this source is $O(m^{-2})$.*

4 Wonham Filter Error Propagation

Let $e_t := p_t - \tilde{p}_t$ denote the filter error process.

Proposition 4.1 (Filter error propagation)**.** *Suppose $\left\|\tilde{\theta} - \theta^*\right\| \le \delta$ for some $\delta > 0$, and that the induced Sharpe ratio errors satisfy $|\delta\rho_i| \le C_\rho \delta$ for some constant $C_\rho > 0$. Also assume $p_0 = \tilde{p}_0$ (both filters initialized identically). Assume further the* filter stability condition

$$\tilde{\lambda}_1 + \tilde{\lambda}_2 \;>\; \tfrac{1}{2}\big(1 + 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2\big), \tag{7}$$

which ensures mean-square contraction of the filter error process. Then there exists a constant $C_F > 0$ depending only on $\bar{\rho}$, $|\rho_1^* - \rho_2^*|$, and T, such that:

$$\sup_{t\in[0,T]} \mathbb{E}\big[e_t^2\big] \;\le\; \frac{C_F\,\delta^2}{\tilde{\lambda}_1 + \tilde{\lambda}_2}. \tag{8}$$

Proof. From (1) and (2), the error e_t satisfies:

$$de_t = \big[b(p_t;\theta^*) - b(\tilde{p}_t;\tilde{\theta})\big]\,dt + \big[a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta})\big]\,d\hat{B}_t,$$

where $b(p;\theta) = -(\lambda_1+\lambda_2)p + \lambda_2$ and $a(p;\theta) = (\rho_1 - \rho_2)p(1-p)$ denote the drift and diffusion coefficients of the Wonham SDE respectively.

With $\delta\lambda_i = \tilde{\lambda}_i - \lambda_i^*$ as defined in Section 2, write the drift difference as:

$$b(p_t;\theta^*) - b(\tilde{p}_t;\tilde{\theta}) = -(\tilde{\lambda}_1 + \tilde{\lambda}_2)\,e_t + \underbrace{\big[-\delta\lambda_2 + (\delta\lambda_1 + \delta\lambda_2)\,p_t\big]}_{\xi_t^b}.$$

Since $p_t \in (0,1)$, we have $|\xi_t^b| \le |\delta\lambda_2| + |\delta\lambda_1 + \delta\lambda_2| \le |\delta\lambda_1| + 2|\delta\lambda_2| \le 3\|\tilde{\theta} - \theta^*\| \le 3\delta$.

Write:

$$a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta}) = \underbrace{(\delta\rho_1 - \delta\rho_2)\,p_t(1-p_t)}_{\xi_t^\sigma} + (\tilde{\rho}_1 - \tilde{\rho}_2)\,\underbrace{\big[p_t(1-p_t) - \tilde{p}_t(1-\tilde{p}_t)\big]}_{\text{Lipschitz in } e_t}.$$

Since $p \mapsto p(1-p)$ is Lipschitz on $[0,1]$ with constant 1, the second term is bounded by

$|\tilde{\rho}_1 - \tilde{\rho}_2| \cdot |e_t|$. Moreover, since $\sup_{p\in[0,1]} p(1-p) = 1/4$, we have $|\xi_t^\sigma| \le |\delta\rho_1 - \delta\rho_2|/4 \le C_\rho \delta$, where the final inequality uses $|\delta\rho_i| \le C_\rho \delta$. The constant C_ρ absorbs the dependence of Sharpe ratio error on (μ, σ) parameterization; under the Euclidean norm, errors $\delta\rho_i$ can exceed δ by a factor of order $1/\sigma$, which is absorbed into C_F.

Itô's formula applied to e_t^2 gives:

$$\frac{d}{dt}\mathbb{E}[e_t^2] = -2(\tilde{\lambda}_1 + \tilde{\lambda}_2)\,\mathbb{E}[e_t^2] + 2\,\mathbb{E}[e_t\,\xi_t^b] + \mathbb{E}\big[\big(a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta})\big)^2\big].$$

Applying Young's inequality $2|e_t\xi_t^b| \le e_t^2 + (\xi_t^b)^2$ and bounding the quadratic diffusion term using the Lipschitz property and the bound on ξ_t^σ:

$$\frac{d}{dt}\mathbb{E}[e_t^2] \;\le\; -\alpha\,\mathbb{E}[e_t^2] + C_1'\,\delta^2,$$

where $\alpha := 2(\tilde{\lambda}_1 + \tilde{\lambda}_2) - 1 - 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2$ and C_1' absorbs the constant from the noise terms. Condition (7) ensures $\alpha > 0$, so that mean-reversion dominates the diffusion contribution to the error dynamics; this is the sense in which the condition is sufficient rather than necessary. Solving the resulting differential inequality via Gronwall's lemma with $e_0 = 0$ yields

$$\mathbb{E}[e_t^2] \le \frac{C_1'\delta^2}{\alpha}\left(1 - e^{-\alpha t}\right) \le \frac{C_1'\delta^2}{\alpha}.$$

Since condition (7) implies α is bounded below by a positive multiple of $\tilde{\lambda}_1 + \tilde{\lambda}_2$, we have $1/\alpha \le C/(\tilde{\lambda}_1 + \tilde{\lambda}_2)$ for a bounded constant C absorbed into C_F, yielding bound (8). □

Remark 4.2. *Condition* (7) *is not satisfied for the calibrated parameters of Wu and Li: with* $\lambda_1^* + \lambda_2^* \approx 3.25$ *and* $|\rho_1^* - \rho_2^*| \approx 4.45$*, one obtains* $\alpha \approx -20.3$*, so the condition is violated by an order of magnitude. This reflects a structural tension in equity market settings: The large Sharpe ratio differential that makes regime information valuable is precisely what causes the diffusion term to dominate mean-reversion in the error SDE. Bound* (8) *may still hold through a different argument (see Open Problem 1 in Section 7), but this cannot be concluded from the present proof.*

Remark 4.3. *The bound is* $O(\delta^2)$*, so parameter errors enter* quadratically*: doubling the estimation error quadruples the belief state error bound. As such, the belief state* $\tilde{p}_t$ *can correctly identify bull and bear regimes qualitatively while still inducing a large policy error through the amplification in Proposition 5.1; these are distinct failure modes.*

5 Policy Sensitivity to Belief State Error

The decomposition is not orthogonal: Term (II) depends on $\mathbb{E}[e_t^2]$, which itself is driven by parameter estimation error. As a result, dominance statements are asymptotic rather than exact.

Define the function:

$$\gamma^*(t,p) \;:=\; \rho_2^* + (\rho_1^* - \rho_2^*)p - (\rho_1^* - \rho_2^*)\, p(1-p)\, \frac{f_p(t,p)}{f(t,p)}, \tag{9}$$

so that $m^*(t,x,p) = -(x-\omega)\,\gamma^*(t,p)/\sigma$.

Proposition 5.1 (Policy sensitivity)**.** *There exists a constant $C_\gamma > 0$ depending only on $|\rho_1^* - \rho_2^*|$, C_1, and C_2, such that $\gamma^*(t,\cdot)$ is Lipschitz in p uniformly over $t \in [0,T]$:*

$$|\gamma^*(t,p) - \gamma^*(t,\tilde{p})| \;\le\; C_\gamma\, |p - \tilde{p}|\,, \qquad \forall\, t \in [0,T],\; p,\tilde{p} \in (0,1). \tag{10}$$

Consequently, assuming $X_t - \omega$ and e_t are weakly correlated (so that second moments factor up to constants), the mean-squared policy error satisfies:

$$\mathbb{E}\big[|m^*(t,X_t,p_t) - m^*(t,X_t,\tilde{p}_t)|^2\big] \;\le\; \frac{C_\gamma^2}{\sigma^2}\,\mathbb{E}[(X_t-\omega)^2]\cdot\mathbb{E}[e_t^2]. \tag{11}$$

Proof. We bound $|\partial\gamma^*/\partial p|$ uniformly in t. Differentiating (9):

$$\frac{\partial\gamma^*}{\partial p} = (\rho_1^* - \rho_2^*)\left[1 - \frac{\partial}{\partial p}\left(p(1-p)\,\frac{f_p}{f}\right)\right].$$

Expanding and applying (R2)–(R3):

$$\left|(1-2p)\,\frac{f_p}{f}\right| \le C_1, \quad \left|p(1-p)\,\frac{f_{pp}}{f}\right| \le \frac{C_2}{4}, \quad \left|p(1-p)\,\frac{f_p^2}{f^2}\right| \le \frac{C_1^2}{4},$$

Defining $C_\gamma := |\rho_1^* - \rho_2^*|(1 + C_1 + C_2/4 + C_1^2/4)$ gives $|\partial\gamma^*/\partial p| \le C_\gamma$ uniformly in (t,p), establishing (10) by the mean value theorem. Bound (11) then follows by squaring, taking expectations, and applying (10):

$$\mathbb{E}\big[|m^*(\cdot,p_t) - m^*(\cdot,\tilde{p}_t)|^2\big] = \frac{1}{\sigma^2}\,\mathbb{E}\big[(X_t-\omega)^2\,(\gamma^*(t,p_t) - \gamma^*(t,\tilde{p}_t))^2\big] \;\le\; \frac{C_\gamma^2}{\sigma^2}\,\mathbb{E}\big[(X_t-\omega)^2\,e_t^2\big]\,.$$

Under the stated weak-correlation assumption, formalized as approximate independence of $(X_t - \omega)$ and e_t, we have $\mathbb{E}[(X_t-\omega)^2\,e_t^2] \approx \mathbb{E}[(X_t-\omega)^2]\cdot\mathbb{E}[e_t^2]$, yielding (11). More precisely, if $\mathrm{Cov}((X_t-\omega)^2,\, e_t^2) \le \kappa\,\mathbb{E}[(X_t-\omega)^2]\,\mathbb{E}[e_t^2]$ for some $\kappa \in [0,1)$, the bound holds with an

additional factor $(1+\kappa)$ absorbed into C_γ^2.

□

6 The Combined Decomposition

Theorem 6.1 (Error decomposition)**.** *Let $m \geq 1$ be the polynomial approximation degree, let $\left\|\tilde{\theta} - \theta^*\right\| \leq \delta$, and fix $\varepsilon \in (0, \frac{1}{2})$. For $p_t, \tilde{p}_t \in [\varepsilon, 1-\varepsilon]$ almost surely, the total mean-squared policy error satisfies:*

$$\mathbb{E}\big[|m^*(t, X_t, p_t) - \tilde{m}(t, X_t, \tilde{p}_t)|^2\big] \;\leq\; \underbrace{\frac{A_\varepsilon}{m^2}\,\mathbb{E}[(X_t-\omega)^2]}_{\text{(I) polynomial approx}} + \underbrace{\frac{C_\gamma^2}{\sigma^2}\cdot\frac{C_F\,\delta^2}{\tilde{\lambda}_1+\tilde{\lambda}_2}\cdot\mathbb{E}[(X_t-\omega)^2]}_{\text{(II) filter propagation}}, \quad (12)$$

where $\tilde{m}$ denotes the implemented policy using the degree-m polynomial approximation evaluated at the estimated belief state $\tilde{p}_t$, $A_\varepsilon = 2(K'_\varepsilon)^2/\sigma^2$ with K'_ε the derivative approximation constant from Remark 3.2, and C_γ, C_F are the constants from Propositions 4.1 and 5.1.

Proof. Apply the triangle inequality followed by $(a+b)^2 \leq 2a^2 + 2b^2$:

$$|m^* - \tilde{m}|^2 \;\leq\; 2\underbrace{|m^*(t,x,p_t) - m^*_m(t,x,p_t)|^2}_{\text{approx error at true } p_t} + 2\underbrace{|m^*_m(t,x,p_t) - m^*_m(t,x,\tilde{p}_t)|^2}_{\text{filter error in approx policy}},$$

where m^*_m denotes the policy using the degree-m polynomial approximation to f evaluated at the true belief state. Taking expectations, we bound each term separately. For the first term: since $m^*(t,x,p) = -(x-\omega)\gamma^*(t,p)/\sigma$ and m^*_m replaces f_p/f with $(q_m)_p$ in γ^*, the approximation error satisfies $|m^*(t,x,p_t) - m^*_m(t,x,p_t)| = |x-\omega|\,|(\ln f)_p - (q_m)_p|/\sigma$. By Remark 3.2, $\|(\ln f)_p - (q_m)_p\|_{L^\infty(\mathcal{D}_\varepsilon)} \leq K'_\varepsilon/m$, so $\mathbb{E}[|m^* - m^*_m|^2] \leq (K'_\varepsilon)^2\,\mathbb{E}[(X_t-\omega)^2]/(\sigma^2 m^2)$, giving a contribution of $2(K'_\varepsilon)^2\mathbb{E}[(X_t-\omega)^2]/(\sigma^2 m^2)$ after multiplying by 2; the factor $1/\sigma^2$ is absorbed into A_ε. For the second term, Propositions 4.1 and 5.1 yield the filter propagation bound; the Lipschitz constant of m^*_m in p differs from C_γ by $O(m^{-1})$ corrections absorbed into A_ε. This gives (12) with $A_\varepsilon = 2(K'_\varepsilon)^2/\sigma^2$. □

Corollary 6.2 (Dominant error source)**.** *Term* (II) *dominates term* (I) *in* (12) *when:*

$$\delta^2 \;\gg\; \frac{(\tilde{\lambda}_1+\tilde{\lambda}_2)\,A_\varepsilon}{C_\gamma^2\,C_F\,\sigma^2\,m^2}.$$

For the calibrated parameters $\mu_1 = 0.25$, $\mu_2 = -0.73$, $\sigma = 0.22$, $\lambda_1 = 0.36$, $\lambda_2 = 2.89$ and polynomial degree $m = 2$, the right-hand side is on the order of 10^{-1}, with the precise value

depending on the PDE regularity constants C_1, C_2 from Assumption 2.1, which Wu and Li do not compute explicitly. Whenever parameter estimation errors are on the order of 10–15% *of the parameter scale or larger, filtering error can therefore be expected to dominate polynomial approximation error.*

7 Discussion

Connection to Wu and Li's empirical findings. Corollary 6.2 formalizes Wu and Li's attribution of performance degradation to filtering error. As noted in the introduction, the heuristic estimator produces $\hat{\lambda}_1 = 1.21$ against a true $\lambda_1^* = 0.36$, placing the parameter error well within the regime where filter propagation dominates. The quadratic amplification of filtering error in Term (II) implies that even moderate parameter misspecification induces a significant policy gap.

The role of the transition rate sum. The factor $(\tilde{\lambda}_1 + \tilde{\lambda}_2)^{-1}$ in bound (II) is the natural self-correction timescale of the Wonham filter, in that faster-switching markets are more robust to parameter misestimation, while near-stationary markets amplify it. POEMV should be deployed with particular care in the latter setting, though this does not appear to be the case for the equity markets considered by Wu and Li.

Tightness of the bounds. The $O(m^{-2})$ rate in Proposition 3.1 is sharp given $C^{1,2}$ regularity; boundedness of third-order derivatives of f would yield $O(m^{-3})$ via the Jackson theorem, suggesting a path toward strengthening Wu and Li's Theorem 1 through higher-order PDE regularity arguments.

Open problems and what a complete decomposition requires. The analysis above establishes Terms (I) and (II) of the decomposition under sufficient conditions and leaves two open problems.

Open Problem 1 (Filter stability in the large-$|\rho_1 - \rho_2|$ regime)**.** Proposition 4.1 establishes bound (8) under condition (7), which requires the total transition rate to exceed $\frac{1}{2}(1+2|\tilde{\rho}_1 - \tilde{\rho}_2|^2)$. For the Wu and Li calibration this threshold is approximately 20.3 against a transition rate sum of 3.25. The condition fails because the Gronwall argument bounds the diffusion coefficient of the error SDE uniformly by $|\tilde{\rho}_1 - \tilde{\rho}_2|$, which is large in equity markets with substantial Sharpe ratio differentials. A complete filter error bound valid for this calibration would require an argument that exploits the geometric structure of the error dynamics more carefully than a uniform diffusion bound permits.

Open Problem 2 (SGD error under a moving loss landscape)**.** Standard finite-time SGD bounds require the loss landscape to be stationary. Here the martingale loss depends on $\tilde{p}_t$, which evolves stochastically and is contaminated by the same parameter error driving Term (II). Consecutive gradient iterates are therefore evaluated at incomparable points, and because both the wealth process and the filter are driven by the same innovation $\hat{B}_t$, the gradient estimator carries a bias proportional to e_t that does not factor out of the decomposition. A formal Term (III) bound would require an SGD analysis that accounts simultaneously for the non-stationarity of the loss landscape and the correlation between gradient bias and filter error.

The two open problems are related: Resolving Open Problem 1 would directly supply the $\mathbb{E}[e_t^2]$ bound needed by the adversarial perturbation route to Open Problem 2. As a sanity check, the gap between the finite-difference optimum (0.067) and POEMV under true parameters (0.083) jointly bounds Terms (I) and (III); its small magnitude suggests their combined contribution is modest relative to the filter propagation term even without formal bounds.

References

Nikolai Vladimirovich Krylov. *Controlled Diffusion Processes*, volume 14. Springer Science & Business Media, 2008.

Bo Wu and Lingfei Li. Reinforcement learning for continuous-time mean-variance portfolio selection in a regime-switching market. *Journal of Economic Dynamics and Control*, 158: 104787, 2024.